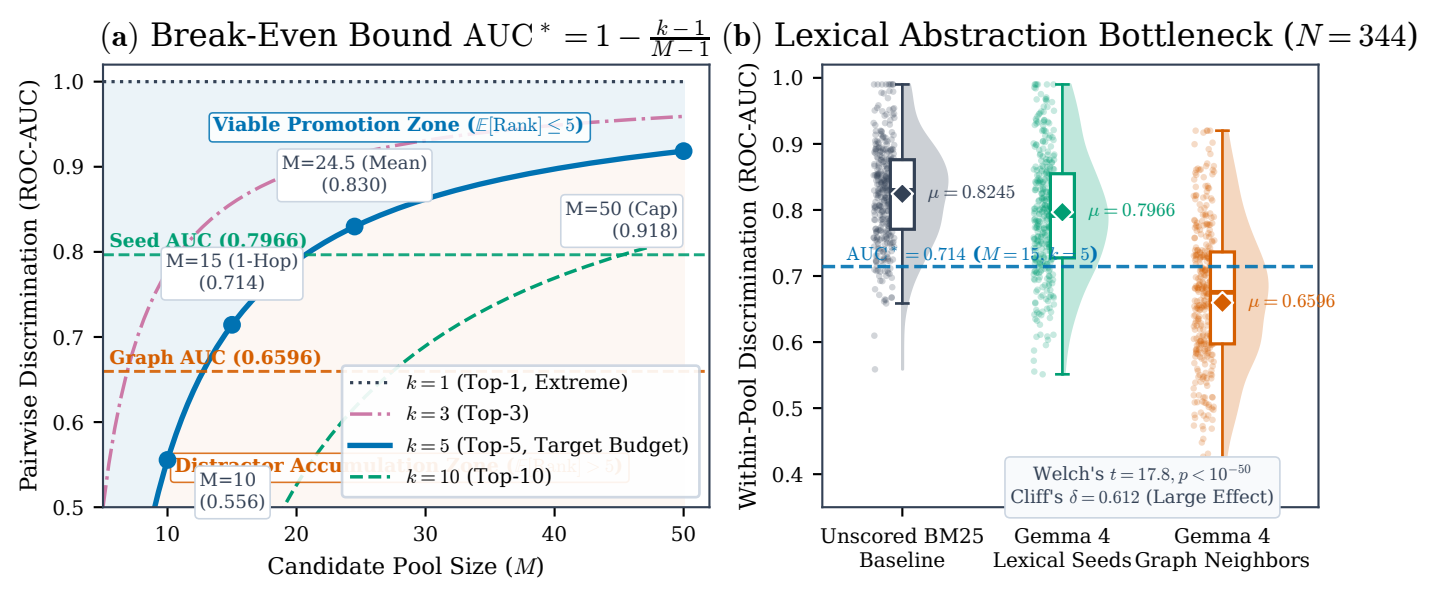

(a) Break-Even Bound AUC* = 1 − (k−1)/(M−1)
(b) Lexical Abstraction Bottleneck (N = 344)
Pairwise Discrimination (ROC-AUC)
Candidate Pool Size (M)
Viable Promotion Zone (E[Rank] ≤ 5)
M=24.5 (Mean)
(0.830)
M=50 (Cap)
(0.918)
Seed AUC (0.7966)
M=15 (1-Hop)
(0.714)
Graph AUC (0.6596)
M=10
(0.556)
k = 1 (Top-1, Extreme)
k = 3 (Top-3)
k = 5 (Top-5, Target Budget)
k = 10 (Top-10)
Within-Pool Discrimination (ROC-AUC)
μ = 0.8245
μ = 0.7966
μ = 0.6596
Welch's t = 17.8, p < 10^−50
Cliff's δ = 0.612 (Large Effect)
Unscored BM25 Baseline
Gemma 4 Lexical Seeds
Gemma 4 Graph Neighbors